

AIRLINE RESERVATION SYSTEM REPORT

1. Introduction

This project is a console-based Airline Reservation and Flight Management System developed in C++ using Object-Oriented Programming principles. The system is designed to simulate real-world airline operations such as flight management, passenger registration, ticket booking, cancellation, and reporting.

2. Objectives

- To apply OOP concepts in a real-world scenario
- To implement inheritance, polymorphism, and abstraction
- To manage airline operations using classes and objects
- To handle errors using exception handling
- To store and retrieve data using file handling

3. OOP Concepts Used

Encapsulation:

All data members are kept private or protected and accessed through methods.

Inheritance:

Flight is a base class, and DomesticFlight is a derived class. Passenger also has multiple derived classes.

Polymorphism:

Functions like calculateBaseFare() and displayDetails() are overridden in derived classes.

Abstraction:

Flight class is abstract and contains pure virtual functions.

4. System Modules

- Flight Management Module
- Passenger Management Module

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- Ticket Booking Module
- Cancellation Module
- Reporting Module
- File Storage Module

5. File Handling

The system uses file handling to store flights, passengers, and tickets. Data is saved when the program exits and loaded when the program starts.

6. Exception Handling

Custom exceptions are used:

- FlightFullException: when no seats are available
- InvalidBookingException: when invalid data is entered

7. UML Class Diagram

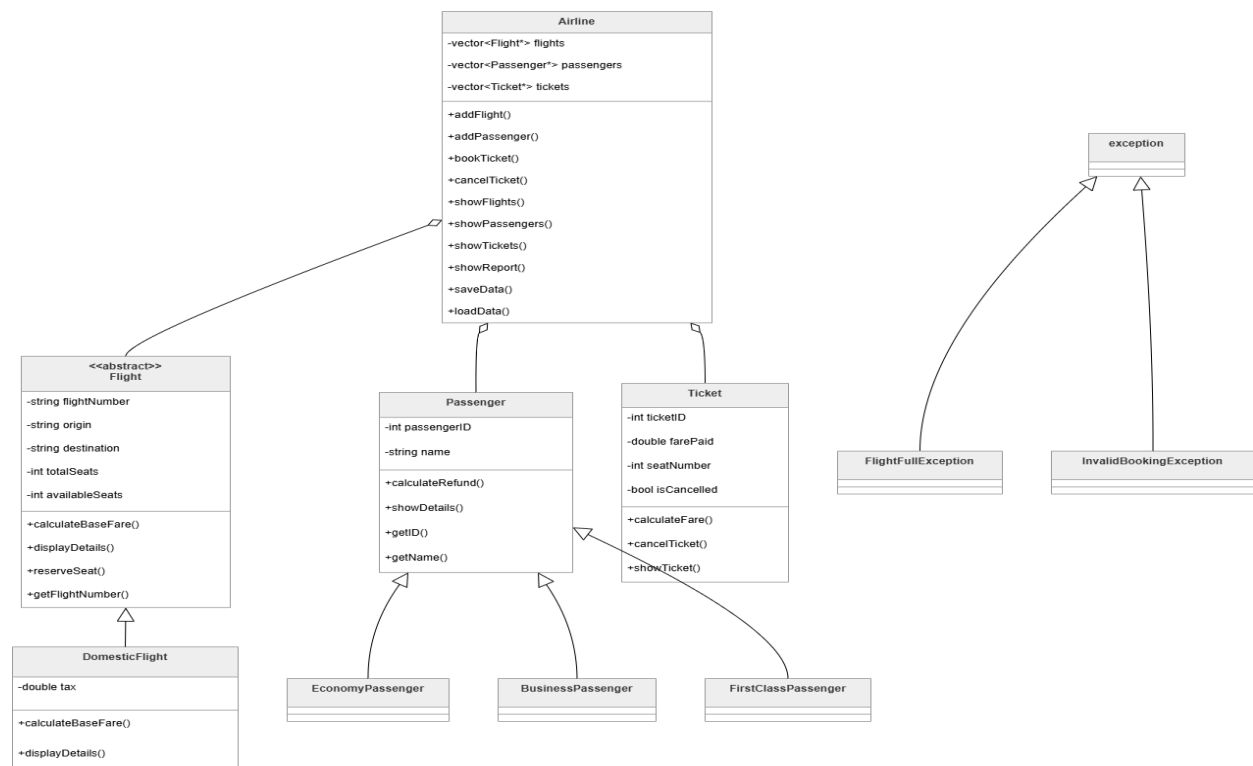


Figure 1: UML Class Diagram of Airline Reservation System

8. Conclusion

This project demonstrates the practical implementation of Object-Oriented Programming in C++. It improves understanding of classes, inheritance, polymorphism, file handling, and exception handling. The system can be further improved by adding a graphical interface and database integration.